

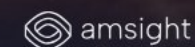
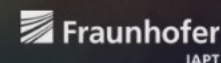
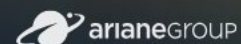
Materials built for the extreme.

Designed with AI. Made and tested in Europe.

FUNDED BY THE EUROPEAN SPACE AGENCY

Initial deployment of PRISM, for its first use cases: ESA
FLPP, FIRST! Simulation & Intelligence.

PRISM ALPHA, WITH



Vulcan 2 on test stand P5, DLR Lampoldshausen. Photo: DLR, CC BY 3.0.

THE PROBLEM

Far too many possible alloys to make them all.

474,311,376

possible alloys from five of nine high-melting metals, mixed in 1% steps

130,000 years

to make each one once, at ten a day

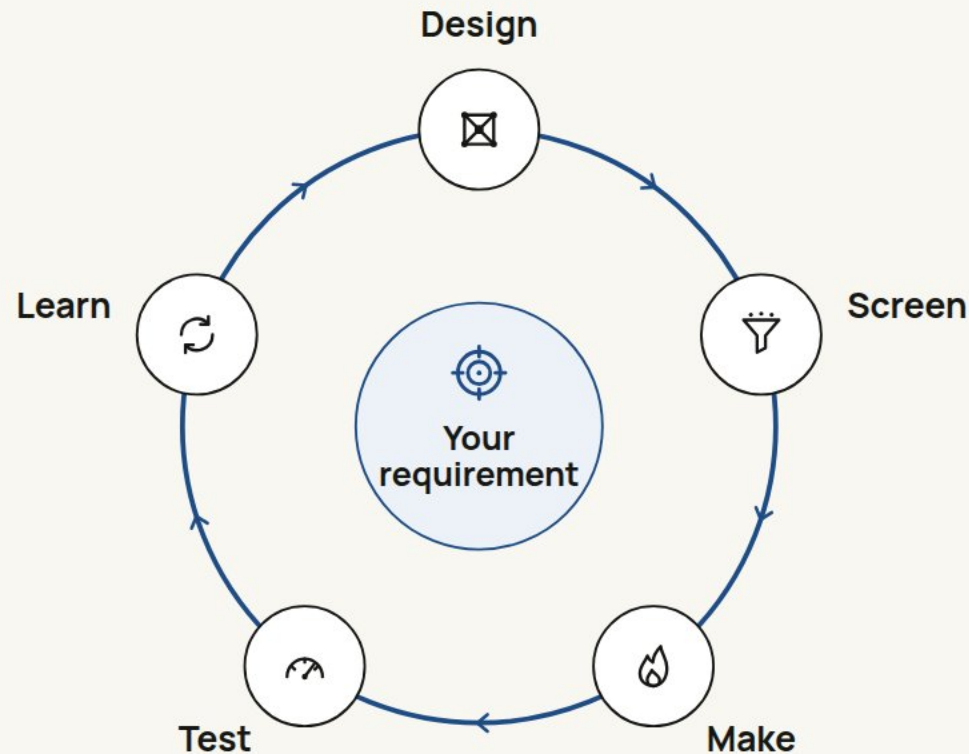
10–20 years

for a new material to reach service today

Sources: Arithmetic: $126 \text{ ways to pick five of nine metals that melt above } 1,650^\circ\text{C} \times 3,764,376 \text{ mixes in whole percent}$. The field is to scale; its layout is illustrative.



One loop: design, make, test, learn.

**Requirement**

You tell us what the part must survive.

PROTOTYPE

**Design**

AI suggests new mixes, from the whole range.

PROTOTYPE

**Screen**

Simulations rule most out, before anything is melted.

PROTOTYPE

**Make**

We melt and 3D-print the best ideas.

IN USE

**Test**

Each sample is measured against your targets.

IN USE

**Learn**

Every result goes back into the models.

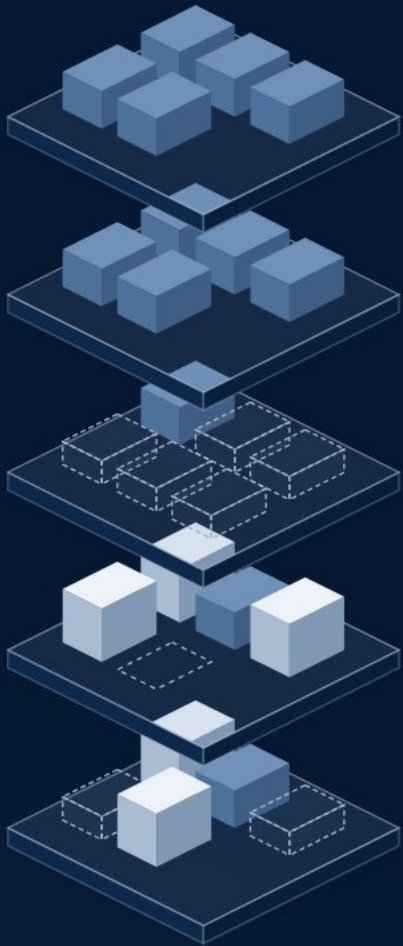
PROTOTYPE

FIRST TARGET

Refractory high-entropy alloys for oxygen-rich preburners in rocket engines, to replace Monel K500 and other legacy alloys.

Five stacks, one system.

■ In use ■ Prototype □ In development --- Goal



01 Research

AI suggests mixes; physics simulations throw out what cannot work.

PROTOTYPE

02 Harness

Plans each round, runs the tools, scores the results, remembers failures.

PROTOTYPE

03 Autonomy

Robots weigh, melt and measure. People stay in charge.

IN DEVELOPMENT

04 Manufacturing and test

Melting, metal 3D printing and testing, with Bimo Tech and Fraunhofer IAPT.

IN USE

05 Evidence

Every result keeps its source, its owner and its rules.

IN USE

Maturity: Each block on a plate is one component, as tall as it is built. In use: runs in current programmes. Prototype: PRISM software, being taken from TRL 3 to 4 in PRISM Alpha. In development: being built.

Four modules, one loop.



Loop: The planner sends a campaign; the samplers propose candidates; simulation and the lab test them; results are kept, with sources; what worked and what failed plans the next round. An engineer signs off what leaves.

Physical synthesis supplies the reward signal the models train on.

01 Recipe translation

PRISM outputs target recipes: precursors, temperatures and thermal profiles.

02 Robotic synthesis

Robot arms dose powder and move samples through heating profiles.

03 Automated characterisation

Diffraction patterns are captured from every synthesised sample.

04 Phase identification

Neural networks compare patterns against structure databases to confirm phases.

05 Feedback

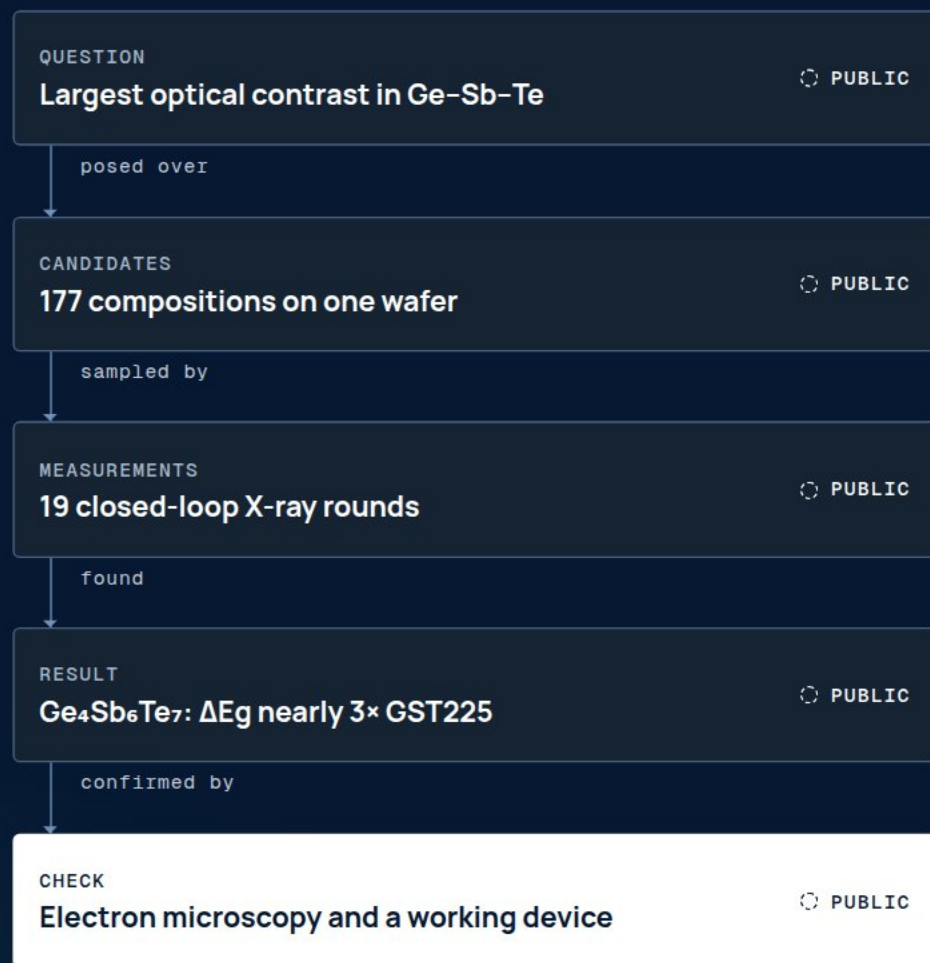
Structured results return to the Evolver's reflector and change the next campaign.

Maturity: In development. Direct instrument control closes the loop in phase III of the plan.



The melt spinner in the materials lab at WUST, Wrocław, where our alloys are melted. We rent the equipment.

Every result carries its own proof.



Partners share data only when they keep control of it. Provenance and rights travel with every result, and derived results inherit the strictest rights of their inputs.

■ PRIVATE

Stays with the organisation that produced it.

▣ COMPUTABLE

Approved workloads compute on it; nobody inspects it.

○ RELEASED

An approved derivative, shared with named parties.

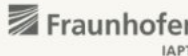
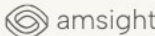
⦿ PUBLIC

Deliberately published.

IP: generative sampling conditioned on thermodynamic feasibility reaches metastable alloys that equilibrium screening excludes. Patent filings in progress.

Maturity: Provenance and export classification maintained on ESA work today; machine-enforced rights in development. Lineage shown: NIST's CAMEO, Kusne et al., Nature Communications 11, 5966 (2020), published work by NIST and partners, not PRISM's.

Funded, contracted and in the lab.

PROGRAMME	STATUS	WHAT EXISTS TODAY		
Project SPARK ESA activity · Bimo Tech	ACTIVE	Eight refractory high-entropy alloy candidates taken to two physical down-selections.		
PRISM Alpha ESA FLPP · FIRST! Simulation & Intelligence	RUNNING	12 months, TRL 3 to 4: at least three candidates and one complete closed loop, with ArianeGroup, Fraunhofer IAPT and amsight.		
PFAS-free polymers Industrial partner · under NDA	CONTRACTED	First privately funded programme and first polymer class. Contract and NDA signed; start pending.		
Fusion heritage Bimo Tech · ITER	DELIVERED	Titanium first-wall materials and rhodium targets supplied to ITER.		
PRISM ALPHA CONSORTIUM				
 Customer · FLPP	 Prime contractor	 Requirements and validation	 LPBF process	 Manufacturing data

Sources: ESA contract records; Mirdyne and Bimo Tech programme records.

Where PRISM goes first.

01 · NOW



Space propulsion

Alloys for the hottest parts of rocket engines. First with ESA and ArianeGroup.

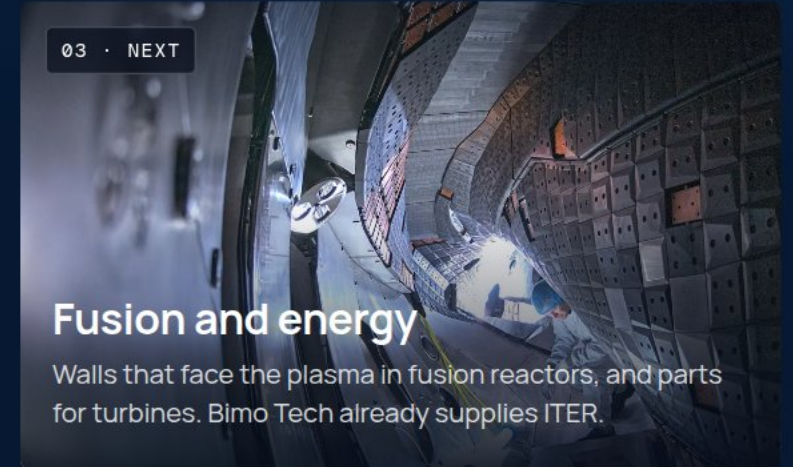
02 · NEXT



Defence

Hot-section materials for aircraft engines, with export rules built in.

03 · NEXT



Fusion and energy

Walls that face the plasma in fusion reactors, and parts for turbines. Bimo Tech already supplies ITER.

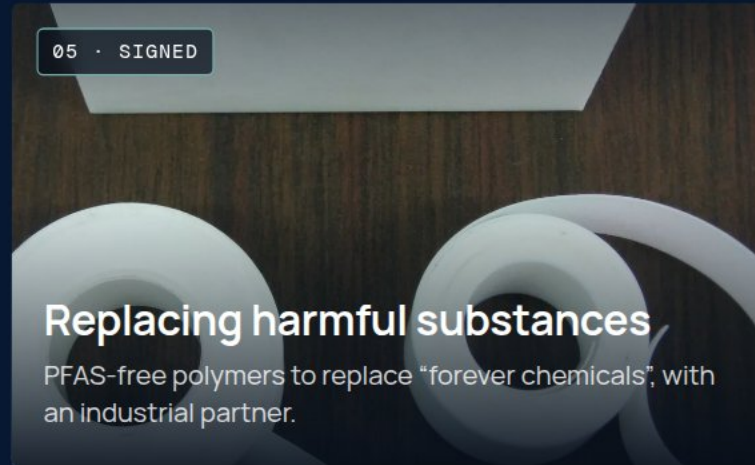
04 · NEXT



Supply independence

Alloys that do the job with metals Europe can get.

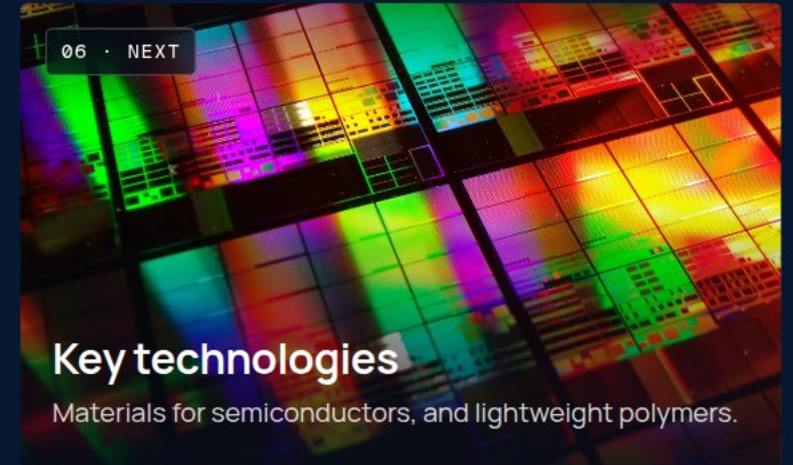
05 · SIGNED



Replacing harmful substances

PFAS-free polymers to replace "forever chemicals", with an industrial partner.

06 · NEXT



Key technologies

Materials for semiconductors, and lightweight polymers.

Photos: Aestus engine in test stand P4.2: DLR, CC BY 3.0 · EJ200 afterburner: Julian Herzog, CC BY 4.0 · Wendelstein 7-X wall tiles: Christopher Roux, EUROfusion, CC BY 4.0 · Tungsten crystals: Alchemist-hp (pse-mendelejew.de), Free Art License. All cropped.

€48M

Serviceable annual wallet

Base case: the upper bound on annual spend across reachable accounts.

€170.7B

EU aerospace
manufacturing
turnover
MKT3

€8.84B

European
space
manufacturing
final sales
MKT4

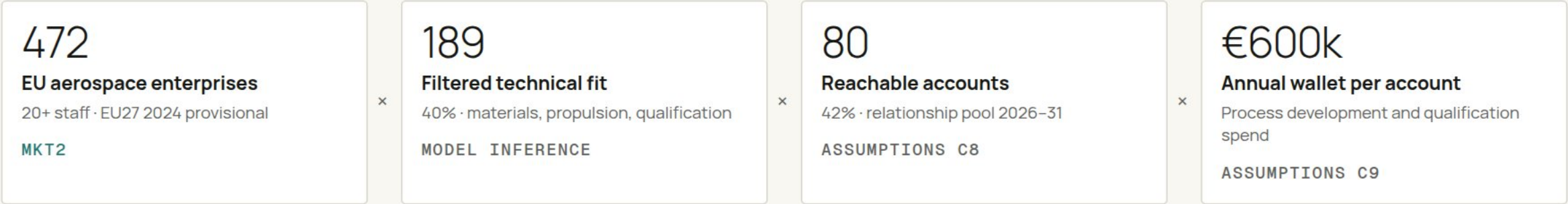
€8.26B

ESA annual
budget, 2026
MKT6

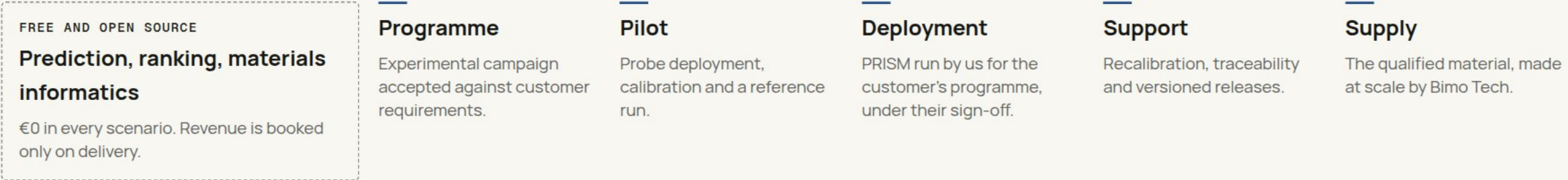
69.6%

Public-
customer
share of space
sales
MKT5

SERVICEABLE ACCOUNT BRIDGE

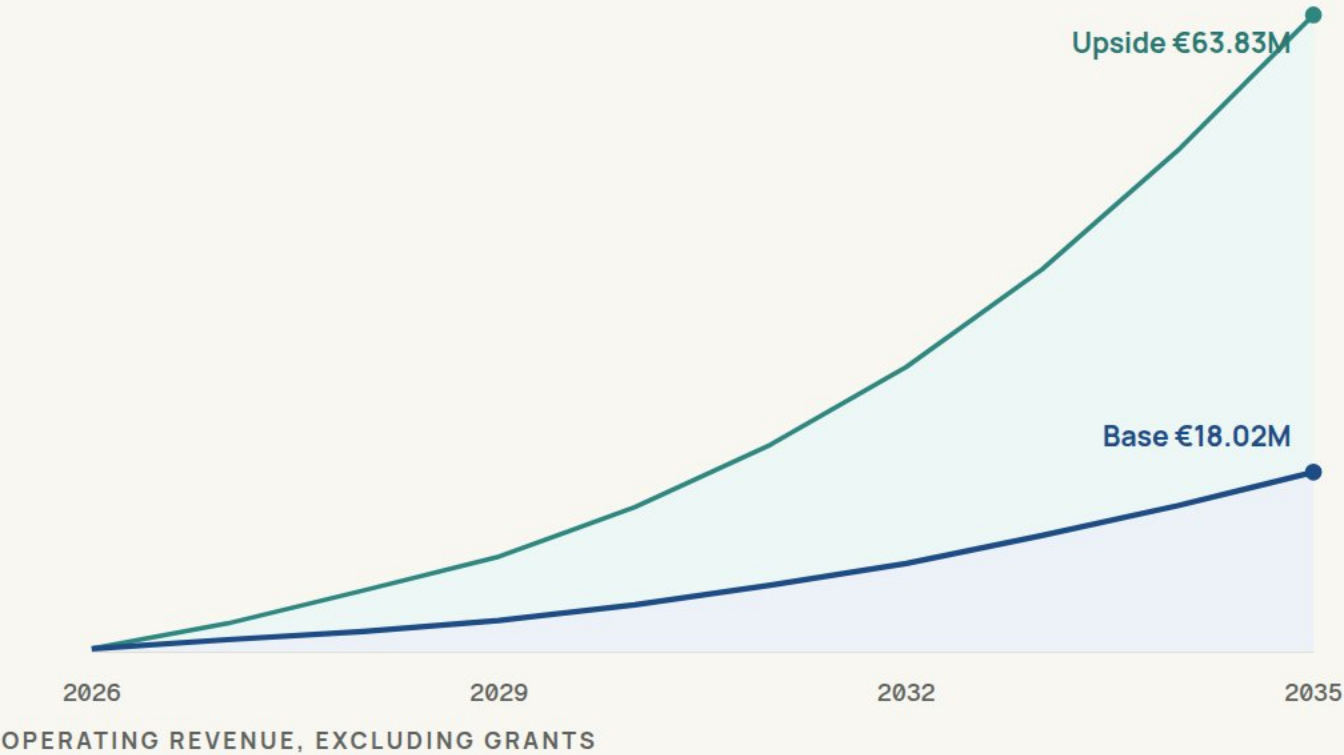


MONETISATION BOUNDARY



Base case: €18.0M revenue by 2035 · EBITDA-positive from 2028 · 61% gross margin. External figures carry their source IDs from the model's evidence register.

Ten-year operating plan.



	BASE	UPSIDE
Operating revenue 2035	€18.02M	€63.83M
EBITDA 2035, before grants	€4.71M	€40.84M
Gross margin 2035	61%	81%
EBITDA-positive from	2028	2027
Cumulative equity required	€2.91M	€0.58M

Revenue is capacity-constrained in both cases: programmes are recognised only up to the delivery capacity left after pilots, deployments and support, so the hiring plan sets the growth rate.

Source: PRISM financial model, research refresh 27 Jul 2026 · checks pass · € nominal.

The team that runs PRISM's ESA work.

KG

Kevin Grüning

MANAGING DIRECTOR, MIRDYNE

Space Systems Lead at Bimo Tech. Physics and technology for space applications, JLU Giessen and THM.

SK

Siddhartha Yash Kovid

TECHNICAL LEAD, MIRDYNE

Technical lead of the ESA projects SPARK and PRISM Alpha at Bimo Tech. Applied AI and data science, MIT Professional Education; biomedical engineering, THM.

MO

Marcin Orzechowski

CO-FOUNDER, MIRDYNE

CEO and Head of R&D at Bimo Tech, which supplies special metals and precision parts for space, energy and science. Wrocław University of Technology.

**KI-Sonderpreis, Hessen Ideen 2026**

The special prize for artificial intelligence. Photo: Hessen Ideen.



Mirdyne is a spin-off of Bimo Tech, which makes special metals and precision parts and supplies ITER.

€4M

Seed round, closing 15 November 2026.

The plan needs €2.91M never to run out of cash. We are raising €4M so the plan survives being wrong, and so the downside gate closes before the money does.

ALLOCATION

Team 65%

Delivery, materials science, product and security headcount

Go-to-market, IP and export control 16%

Bids, consortia, field selling, patents, dual-use compliance

Compute, infrastructure and lab access 13%

Controlled core, per-site secure compute, partner-lab access

Probe hardware, tools and testing 6%

Probe builds, calibration, R&D tooling

ROADMAP · 24 MONTHS

- I **Computational validation**
6 months. The full discovery loop running the first PRISM campaigns.
- II **Hybrid loop**
12 months. Integration with Fraunhofer IAPT additive-manufacturing data.
- III **Autonomous laboratory**
24 months. Direct instrument control closes the loop with no human in the sequence.

MIRDYNE · GIESSEN, GERMANY

Start with the capability you need.

Alloys and polymers, designed, made and tested in one loop, with the proof attached.

Register interest →

mirdyne.com →

Initial deployment of PRISM, for its first use cases, funded under ESA FLPP, FIRST! Simulation & Intelligence.

Mirdyne is a spin-off of Bimo Tech. [Technology concept by marc27](#)

